

Charalampos Lazaridis Patsalias

MSc AI ů sequence models, training objectives, interpretability

✉ mpampis.laz@gmail.com

🌐 clazaridis.com

🐙 Babis1987

in charalampos-lazaridis-patsalias

Profile

ML researcher with five years in industry (biochemical ML, LLM systems) and an MSc thesis on how architecture and training objective shape the representations sequence models learn.

Education

2024–2026 **MSc Artificial Intelligence, University of Essex**

Thesis (October 2026): *The Emergence of Internal Representations in Sequence Models: Comparison of Architectures and Objectives*, supervised by Athanasios Chasapis. Factorial study (Transformer vs Mamba $\times$ autoregressive vs JEPA $\times$ three board sizes, 12 conditions, 5 seeds). The objective's effect reverses between architectures (+5.09 pp interaction at 16 $\times$ 16, 95% CI 4.23–5.95). Causal interventions beat a matched random control in all 12 conditions.

2012–2016 **MSc Theoretical Physics, Utrecht University, Netherlands**

General relativity, string theory, group theory, statistical physics. Thesis combining statistical physics with game theory.

2006–2011 **BSc Physics, Aristotle University of Thessaloniki, Greece**

Mathematics, probability, statistics, modelling, numerical reasoning.

2017–2018 **One year of BSc coursework, Data Science & Knowledge Engineering, Maastricht University, Netherlands**

Programming from zero: algorithms, discrete mathematics, data science. Teaching assistant in Mathematical Analysis. Left for the GLI role.

Industry research

Jan 2022 – Aug 2025 **ML & AI Researcher, biotech startup**

- Cut a candidate list from thousands of compounds to 50 by combining LLM extraction over 100k+ PubMed papers (validated with LLM-as-a-judge), Lipinski filtering and AutoDock Vina docking; the 3rd-ranked compound was selected for development after lab testing.
- Built a bioavailability classifier (ensemble over more than 1,800 molecular descriptors, reduced with Elastic Net) reaching ROC-AUC 0.85+; its ranking decided which compounds went to the lab.
- Cut report generation from 15 minutes to 2 minutes by replacing sequential LLM calls with asynchronous execution.
- Automated training and inference for 616 toxicological models for systematic compound-safety assessment, alongside ADME property models.
- Built a LangGraph agent with vector stores, tool calling (search engines, research databases) and intent-based routing.

Oct 2018 – Jul 2020 **Math Analyst, GLI Europe, Haarlem, Netherlands**

Certified the statistical fairness of 100+ games a year for regulators in several jurisdictions, running and validating C++ Monte Carlo simulations.

2020–2021 **Retraining into data science, online courses and certificates**

Systematic coursework in data science and predictive modelling, closing with the IBM Data Science capstone (Dec 2021).

Selected projects

Emotion & image app	Flask app for face detection, 7-emotion classification with a CPU-only mini-Xception, age/gender estimation and image description with Mistral-7B. Live demo on Hugging Face Spaces. Demo ũ Code
Customer churn	MLP (Optuna-tuned) benchmarked against logistic regression and CatBoost on the Telco dataset; class weights beat SMOTE; SHAP analysis of the best model (F1 0.625, ROC-AUC 0.836). Report (38 pp.)

Skills

ML stack	Python, PyTorch, TensorFlow/Keras, scikit-learn, CatBoost
Research	Transformers, Mamba/SSMs, JEPA; linear/MLP probing, causal interventions; reproducible experiments (SHA-256-linked checkpoints, data manifests)
LLM systems	LangChain/LangGraph, RAG, LLM-as-a-judge
Other	SQL, C++ (working knowledge), Docker (basic); RDKit, Mordred, AutoDock Vina

Languages

Greek (native), English (C1+)